

WEEK 1- REPORT

Research and Planning for E-Governance Data Initiatives

Topic: Data Analysis Strategy for UMANG App

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TABLE OF CONTENTS

Sr.No.	Contents	Page No.
1.	Introduction.....	3
2.	Objectives of the Study.....	4
3.	Overview of UMANG App.....	5
4	Why I Chose UMANG App for This Research.....	7
5.	Personal Research and Observations.....	8
6.	Key Data Sources.....	11
7.	Key Metrics for Analysis.....	12
8.	Data Collection Methods.....	13
9.	Analytical Techniques.....	14
10.	Proposed System Architecture.....	15
11.	Practical Challenges Observed.....	16
12.	Suggested Improvements.....	17
13.	Timeline for Implementation.....	18
14.	Expected Outcomes.....	19
15.	Flowchart.....	20
16.	Conclusion.....	22
17.	Submission Description.....	23
18.	References.....	25

1. Introduction

E-Governance refers to the use of digital technologies by governments to provide services, exchange information, and improve communication with citizens. India has rapidly expanded digital governance initiatives through platforms that connect multiple public services into unified systems.

One such platform is the UMANG App. It is a centralized government application that provides access to various public services such as EPFO, DigiLocker, Aadhaar-related services, bill payments, government schemes, certificates, and many more through a single application.

The purpose of this report is to analyze how data generated through UMANG can help improve digital governance systems, identify practical challenges faced by users, and support better decision-making through data analysis techniques.

2. Objectives of the Study.

The objectives of this project are:

- To study the role of data in digital governance systems.
- To analyze how the UMANG App provides multiple government services through one platform.
- To identify practical challenges faced by users.
- To understand how user data and service data can improve decision-making.
- To suggest improvements for better accessibility and efficiency.
- To propose a data analysis strategy for improving digital public services.

3. Overview of UMANG App

UMANG App is a digital governance platform developed under the Digital India initiative. The application integrates multiple government departments and services into a single mobile platform.

The app provides services related to:

- EPFO
- DigiLocker
- Aadhaar
- PAN services
- Utility bill payments
- Government schemes
- Job and skill development services
- Farmer-related schemes
- Health and education services
- Birth and other certificates

The application aims to reduce the need for citizens to visit multiple government offices or websites separately

MAJOR SERVICES AVAILABLE ON UMANG APP



Aadhaar
Services



EPFO
Services



DigiLocker



PAN
Services



Bill
Payments



Government
Schemes



Farmers
Services



Job & Skill
Services



Health
Services



Certificates



Education
Services



Police
Services



Pension
Services



Loans &
Finance



More
Services

4. Why I Chose UMANG App for This Research

Many people usually choose commonly discussed topics or depend only on information already available on the internet for completing research projects. However, I wanted to select a platform that people often ignore despite it being highly useful in real life.

I chose the UMANG App because it combines many government services into one platform and provides a better opportunity to understand how digital governance works practically.

Instead of only collecting theoretical information, I personally explored the application to observe how users interact with the services available inside the platform. This helped me gain a more practical understanding of the application and made the research more realistic rather than completely opinion-based.

5. Personal Research and Observations

During this project, I explored the UMANG App personally to understand its working and usability.

After opening the application, I observed that the front page was user-friendly and simple to understand. The app also provided easy login methods such as OTP-based login, making access easier for users.

One interesting feature I noticed was a popup notification regarding the Class 10 board examination results. Since this topic was currently trending, it showed that the application dynamically highlights important public services according to current events and user needs.

While exploring the application further, I found multiple categories of services for different groups such as:

- 1) Farmers
- 2) Students
- 3) Professionals

- 4) Workers
- 5) Job seekers
- 6) Pensioners
- 7) The application also included services related to:
- 8) DigiLocker
- 9) Government schemes
- 10) EPFO passbook services
- 11) Skill development schemes
- 12) Loan-related services
- 13) Aadhaar updates
- 14) Birth certificates
- 15) Other public documents

The services were properly organized and easy to navigate, which makes UMANG a useful digital platform for citizens.

However, I also identified some practical limitations.

While accessing EPFO services, I observed that login access mainly depended on the UAN (Universal Account Number). There appeared to be limited alternative recovery or authentication methods for users who may forget their UAN or lose access to their registered mobile numbers.

Another issue is that the application completely depends on internet connectivity. Users living in rural areas or places with poor network access may face difficulties accessing important services.

As a possible improvement, the application could introduce alternative verification systems such as fingerprint authentication or face recognition through smartphones. This could make access easier and more secure for users facing login difficulties.

Overall, my personal exploration helped me understand how digital governance platforms are designed to provide citizen-centric services while also highlighting areas where accessibility and usability can still be improved.

6. Key Data Sources

The following data sources can be used for analysis:

Internal App Data

Login frequency

Service usage records

Transaction data

Error logs

Government Data

Digital India reports

MeitY reports

UMANG usage statistics

External Sources

User reviews

Feedback forms

Public surveys

System Data

API response time

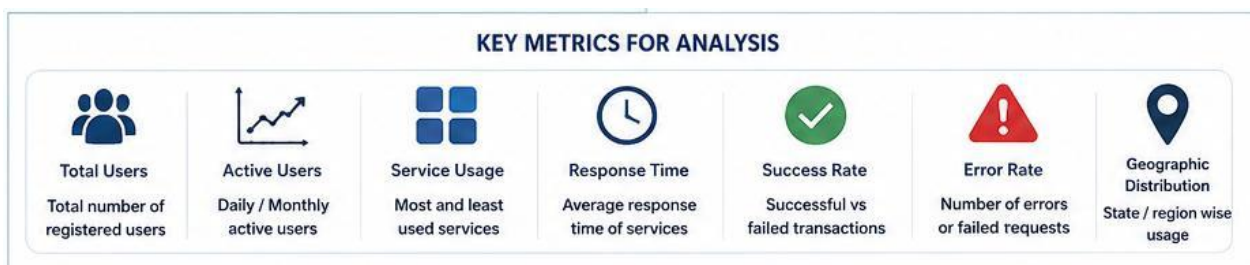
Server load data

Downtime records

7. Key Metrics for Analysis

The following metrics can help evaluate the effectiveness of the UMANG App:

- 1) User Metrics
- 2) Daily active users
- 3) Monthly active users
- 4) User retention rate
- 5) Service Metrics
- 6) Most used services
- 7) Least used services
- 8) Service response time
- 9) Performance Metrics
- 10) System uptime
- 11) Error rates
- 12) Transaction success ratio
- 13) Geographic Metrics
- 14) Rural vs urban usage
- 15) State-wise user distribution



8. Data Collection Methods

The following methods can be used for collecting data:

- Automated Methods
- App usage tracking
- API monitoring
- System log analysis
- Manual Methods
- User surveys
- Feedback forms
- Public reports
- Secondary Research
- Government websites
- Online research papers
- Digital India publications

9. Analytical Techniques

The following analytical techniques can be used:

Descriptive Analysis

To understand user behavior and service usage trends.

Comparative Analysis

To compare the popularity and performance of different services.

Trend Analysis

To observe changes in app usage over time.

Predictive Analysis

To estimate future service demand and server load.

Visualization Techniques

Pie charts

Bar graphs

Heat maps

Dashboards

10. Proposed System Architecture

SYSTEM ARCHITECTURE OF UMANG APP



11. Practical Challenges Observed

During the research and personal usage of the app, the following practical issues were identified:

1. UAN Dependency

EPFO services mainly depend on UAN-based login systems. Users who forget their UAN or lose access to their mobile numbers may face difficulties accessing services.

2. Internet Dependency

The application requires stable internet access. Users from rural or low-network areas may experience difficulties.

3. Awareness Issues

Many citizens are still unaware that multiple government services are available in one application.

4. Navigation Complexity

Because the application contains many services, some users may find it difficult to locate the correct section.

5. OTP Delays

Users may occasionally face delays in OTP verification during login or authentication.

12. Suggested Improvements

1. Alternative Verification Methods

The app should provide alternative login options such as biometric verification or recovery systems for users facing UAN-related issues.

2. Better Categorization

Services should be grouped more clearly for students, farmers, professionals, and senior citizens.

3. Regional Language Support

Adding more regional language support can improve accessibility for rural users.

4. Awareness Campaigns

Government awareness programs can help more citizens learn about UMANG services.

5. Smart Search System

An AI-based search feature could help users quickly locate services.

13. Timeline for Implementation

Phase:

Activity for this projection presentation:

Duration:

1. Phase 1 - Data Collection
1 Week
2. Phase 2 - Data Cleaning
3–4 Days
3. Phase 3 - Data Analysis
1 Week
4. Phase 4 – Visualization
4–5 Days
5. Phase 5 - Final Reporting
3 Days

14. Expected Outcomes

The proposed data analysis strategy may help achieve:

Better service delivery

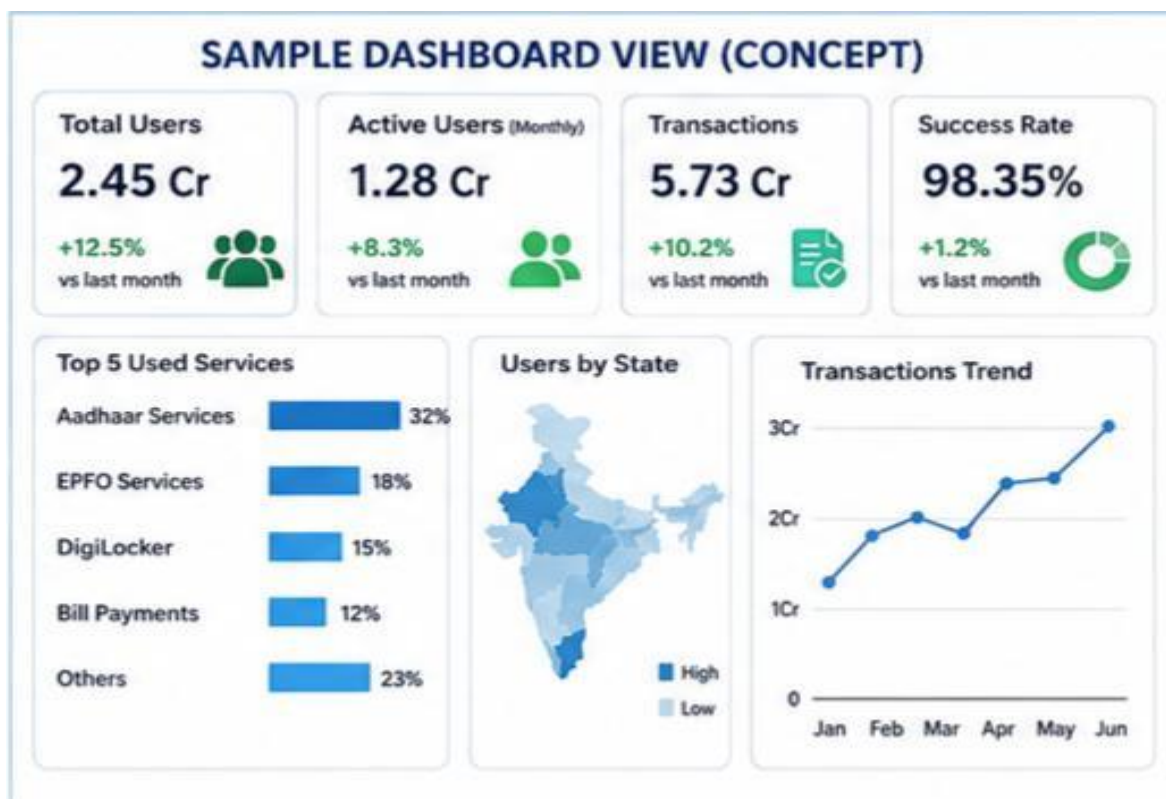
Improved user satisfaction

Faster issue detection

Reduced system downtime

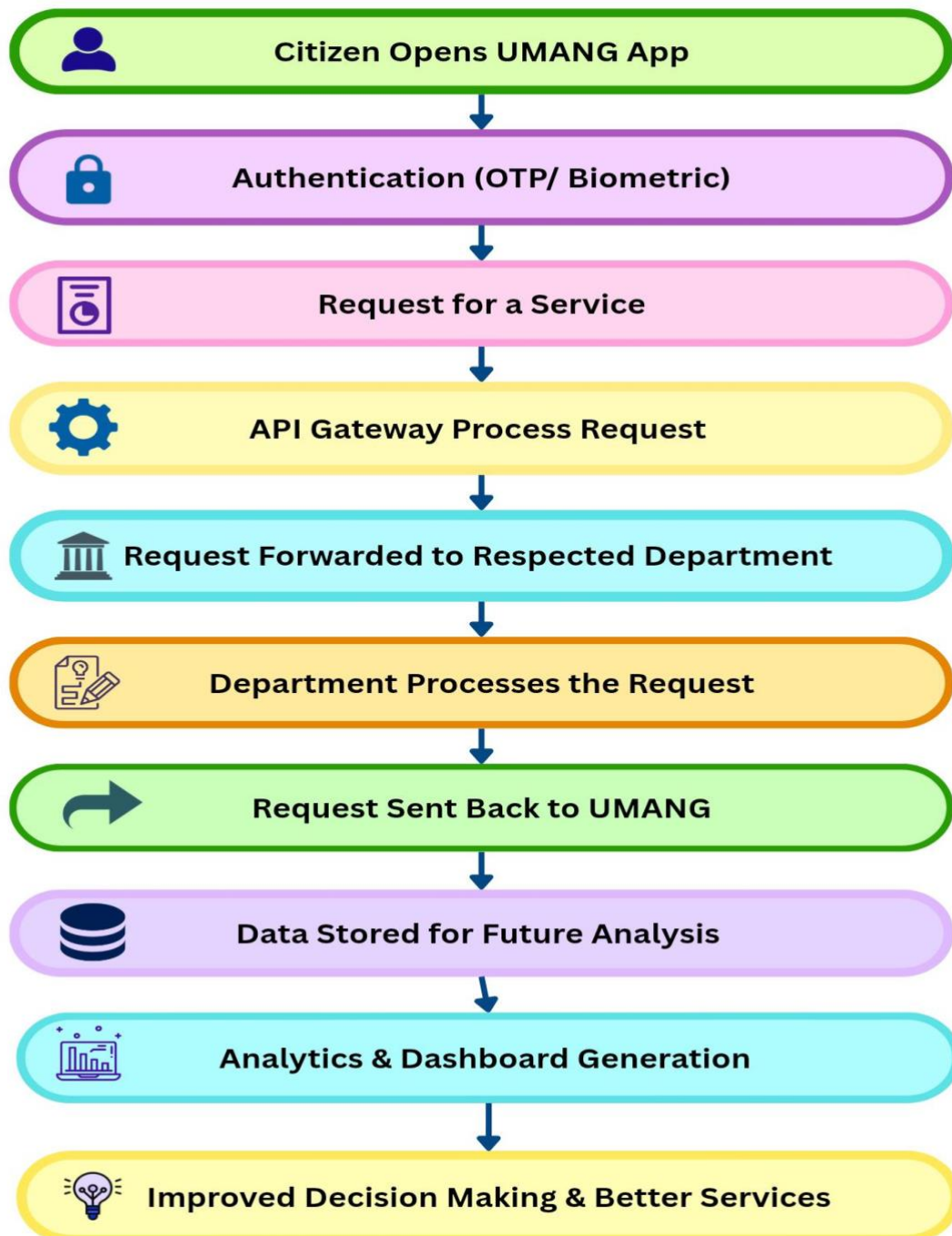
Better understanding of citizen needs

Improved decision-making in digital governance



15. Flowchart

DATA FLOW DIAGRAM



FLOWCHART OF DATA ANALYSIS PROCESS



16. Conclusion

The UMANG App is a strong example of integrated digital governance in India. Through this project, I understood how a single application can provide multiple government services efficiently to citizens.

My personal exploration of the application helped identify both its strengths and practical limitations. While the app provides a wide range of useful services, issues such as UAN dependency, internet connectivity, and user awareness still need improvement.

By applying proper data analysis methods, the government can better understand citizen needs, improve service efficiency, and strengthen digital governance systems in the future.

17. Submission Description

This report presents a detailed research and planning strategy for analyzing the UMANG App as part of India's digital governance ecosystem. The study focuses on how data generated from government digital platforms can help improve public services, decision-making, and citizen accessibility.

The report includes practical observations based on personal exploration of the UMANG application along with theoretical research related to e-governance systems. Multiple services available inside the application such as EPFO, DigiLocker, Aadhaar-related services, government schemes, and certificate services were studied during the research process.

The report also identifies practical challenges such as UAN dependency, internet accessibility issues, OTP verification problems, and awareness limitations among users. Based on these observations, several improvements have been suggested including biometric verification systems, better

categorization of services, regional language support, and smarter search features.

Additionally, the report proposes a structured data analysis strategy including data collection methods, key metrics, analytical techniques, and implementation planning. The overall objective of this research is to demonstrate how data analysis can improve citizen-centric digital governance systems in India.

18. References

UMANG Official Website:

<https://web.umang.gov.in/landing/>

Digital India Initiative:

<https://www.digitalindia.gov.in/initiatives/>

Ministry of Electronics and Information Technology (MeitY):

<https://meity.gov.in/>

National Payments Corporation of India (NPCI):

<https://www.npci.org.in/>

Reserve Bank of India (RBI):

<https://m.rbi.org.in//home.aspx>

My Github:

[Rohan's GitHub Link](#)

<https://github.com/rohanw2004/E--Governance-Digital-Service>